

Problem Set 2

POLS W4732

Fall 2026

Due October 2 before the TA section

Artificial-intelligence policy. Unless an exercise explicitly states otherwise, students may not use generative-AI systems (including ChatGPT, Claude, Gemini, Copilot, or AI-generated search answers) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, typesetting software, the teaching team, and collaboration within the registered problem-set group are permitted. Any approved computational tool must be disclosed. Every group member is responsible for understanding and being able to explain the entire submission.

Exercise 1 (A twice-repeated game). *Consider the stage game*

		Player 2		
		A	B	C
Player 1	A	9, 9	1, 6	-2, 10
	B	6, 1	5, 5	1, 0
	C	10, -2	0, 1	2, 2

Total payoff is $u^1 + \delta u^2$, where $\delta \in (0, 1]$. The proposed symmetric strategy plays A in period 1; after (A, A) it plays B; after any other first-period outcome it mixes between B with probability q and C otherwise.

- 1. Write a separate incentive constraint for every distinct first-period unilateral deviation; do not assume which one binds.*
- 2. Distinguish Nash equilibrium of the repeated game from subgame-perfect equilibrium. Evaluate the claim that the profile is automatically SPE because the continuation at $q = 1/6$ is a stage-game equilibrium.*
- 3. Characterize the set of (δ, q) supporting Nash equilibrium and the subset supporting SPE.*

Exercise 2 (A revenue-sharing agreement). *At date 0, both players simultaneously accept (A) or reject (R) a revenue-sharing agreement. If either rejects, they play the following Prisoner's Dilemma in periods 1 and 2:*

	<i>C</i>	<i>D</i>
<i>C</i>	2, 2	0, 3
<i>D</i>	3, 0	1, 1

If both accept, Player 1 receives share α and Player 2 receives $1 - \alpha$ of the sum of the two stage payoffs in each period. Player i pays enforcement cost k_i at date 0. Use unnormalized discounted payoff $u^1 + \delta u^2$ and subtract k_i once.

1. Describe the terminal histories without listing them and give their number when there are T post-agreement periods.
2. Derive both acceptance constraints and characterize the feasible set of (α, k_1, k_2) .
3. Explain when no division rule induces agreement.

Exercise 3 (Tit-for-Tat). In the infinitely repeated Prisoner's Dilemma above, a proposed solution checks deviations only after cooperative histories and concludes that Tit-for-Tat is an SPE whenever $\delta \geq 1/2$.

1. Identify the missing histories and reconstruct the complete one-shot-deviation analysis.
2. Now let a player who observed D in the previous period retaliate with probability q and play C otherwise, using fresh independent randomization. Write the continuation-value system and characterize the (δ, q) pairs that satisfy sequential rationality after both cooperative and noncooperative histories.

Exercise 4 (Minmax and credible punishment). Consider the infinitely repeated game

		Player 2		
		<i>A</i>	<i>B</i>	<i>C</i>
Player 1	<i>A</i>	4, 4	3, 0	1, 0
	<i>B</i>	0, 3	2, 2	1, 0
	<i>C</i>	0, 1	0, 1	0, 0

Initial play is (B, B) . After every history, the public state is either Reward, Punishment 1, Punishment 2, Punishment 3, or Permanent Punishment. Reward prescribes (B, B) . A deviation starts three periods of (C, C) ; compliance returns to Reward, while any deviation during punishment moves to Permanent Punishment, which prescribes (C, C) forever.

1. Represent this strategy as a finite-state automaton, including every transition.
2. At every state, identify the most profitable one-shot deviation.
3. If it is not an SPE, provide the shortest deviation certificate.

4. Design the closest credible replacement strategy attempting to sustain (B, B) , or prove that no strategy in this class can do so. Explain why a minmax payoff need not be a credible continuation payoff.

Exercise 5 (Alternating offers with breakdown). *Players divide a unit surplus and are perfectly patient. After rejection of Player 1's offer, bargaining ends with payoffs (b_1, b_2) with probability p_1 ; after rejection of Player 2's offer, it ends with the same payoffs with probability p_2 .*

1. Recover the two stationarity equations and solve the unique stationary SPE division.
2. Verify that setting $p_1 = p_2 = p$ gives

$$v_1 = \frac{1 - b_2 + (1 - p)b_1}{2 - p}, \quad v_2 = \frac{1 - b_1 + (1 - p)b_2}{2 - p}.$$

3. Check the limits $p \rightarrow 0$ and $p \rightarrow 1$ and identify which payoff is more sensitive to each asymmetric breakdown probability.

Editorial source note: obsolete exercises that were formerly placed after the first `\end{document}` have been removed from the 2026 source. Remove this note after reviewing the revision.